

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the file

taken Sample data 1% of records from this hour and combined 12 files using for loop

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

Show all columns, their data types, missing values, and number of unique values

2.1.2. Combine the two airport_fee columns

Drop the column [airport_fee", "Airport_fee"]

And rename the airport_fee

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

Using the df.isnull.sum() and sort the values

2.2.2. Handling missing values in passenger_count

Impute the Missing values in Passenger_count using Median values

2.2.3. Handle missing values in RatecodeID

Here Impute the missing values for Most Repeated values for RatecodeID

Using Mode

2.2.4. Impute NaN in congestion_surcharge

Here the impute the missing the values for **congestion_surcharge** for Most repeated values using for Mode value

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

Here using outliers, IQR, Q1, Q2, Q3 and also median values using Box Plot

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

Categorical variable is congestion_surcharge, DOLocationID, PULocationID, passenger_count, payment_type

And Numerical Variable is fare_amount, extra, mta_tax, tip_amount, tolls_amount and total_amount

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and

Hourly trends may show high pickups during rush hours (morning and evening)

Day of the week trends may indicate higher pickups on weekends or weekdays.

Monthly distribution can reveal seasonality (e.g., higher during holidays or tourist seasons).

3.1.3. Filter out the zero/negative values in fares, distance and tips

fare_amount > 0: Filters out free rides or data entry errors and

trip_distance > 0: Ensures valid trips with distance.

tip_amount >= 0: Tips can be zero, but not negative.

4.1.1. Analyse the monthly revenue trends

Are there seasonal spikes in revenue (e.g., holidays, tourist season)

Is the trend increasing or decreasing over time

Any dips that could indicate anomalies or external factor.

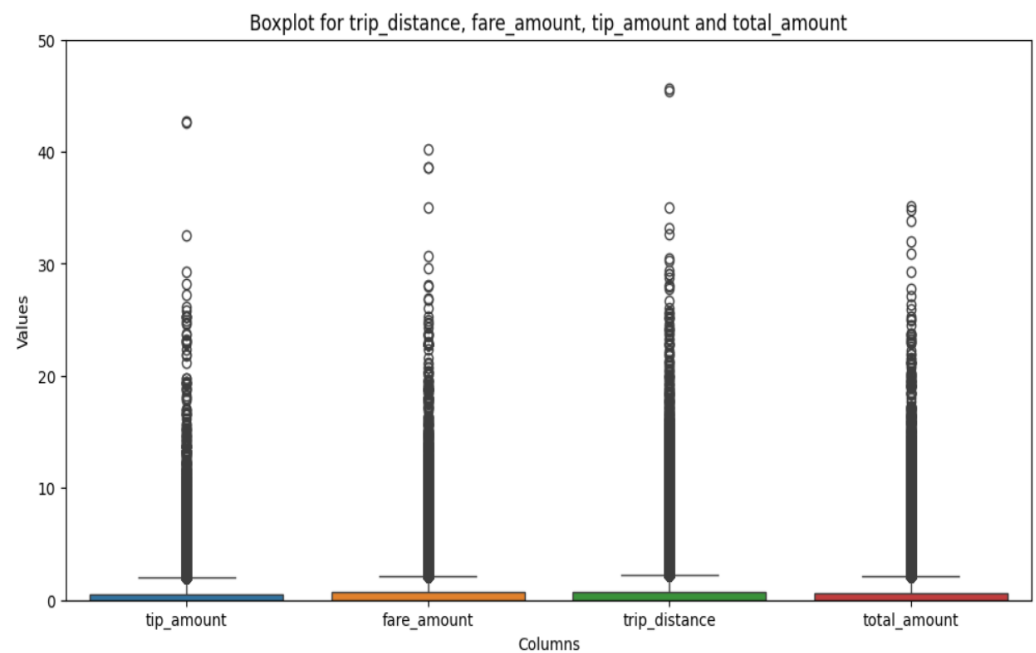
4.1.2. Find the proportion of each quarter's revenue in the yearly revenue

Attached the JPEG Format (Image)

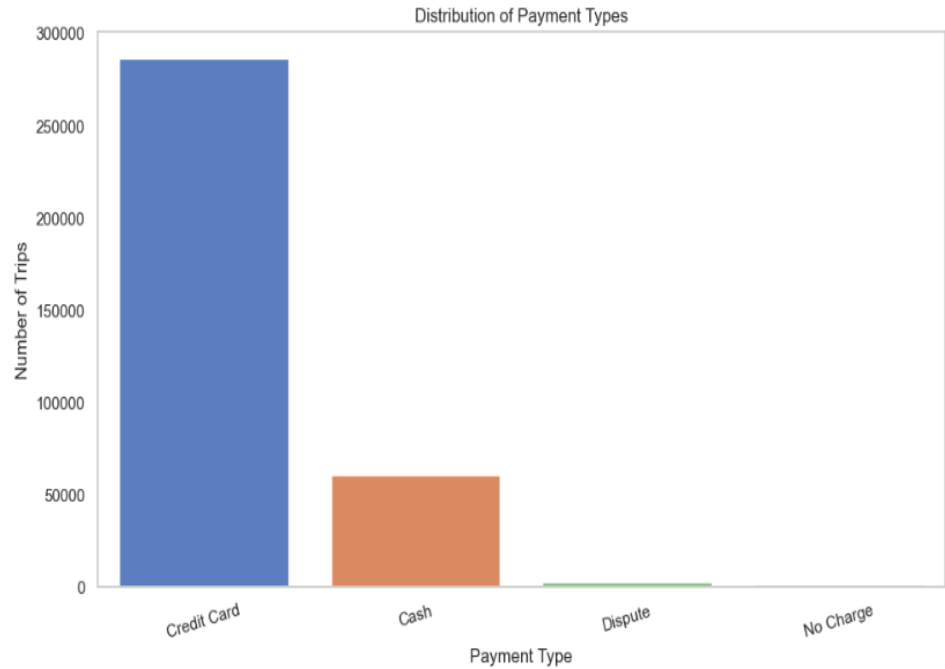
4.1.3. Analyse and visualise the relationship between distance and fare amount

hererAttched Images

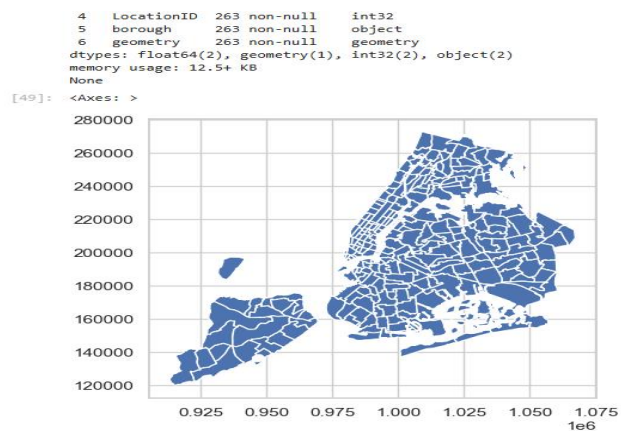
4.1.4. Analyse the relationship between fare/tips and trips/passengers



4.1.5. Analyse the distribution of different payment types



4.1.6. Load the taxi zones shapefile and display it



4.1.7.

```
)}]: # Merge zones and trip records using LocationID and PULocationID
df_geo = df_nonzero.merge(
    zones[['LocationID', 'zone', 'borough']],
    left_on='PULocationID',
    right_on='LocationID',
    how='left'
)
df_geo.head()
```

	VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	RatecodeID	PULocationID	DOLocationID	payment_type	fare_amount	...
0	2	2023-01-01 00:07:18	2023-01-01 00:23:15	1.0	2.641593	1.0	138	256	2	1.500000	...
1	2	2023-01-01 00:16:41	2023-01-01 00:21:46	2.0	-0.234513	1.0	161	237	1	-0.444444	...
2	2	2023-01-01 00:14:03	2023-01-01 00:24:36	3.0	-0.146018	1.0	237	141	2	-0.166667	...
3	2	2023-01-01 00:24:30	2023-01-01 00:29:55	1.0	-0.544248	1.0	143	142	2	-0.555556	...
4	1	2023-01-01 00:42:56	2023-01-01 01:16:33	2.0	2.358407	1.0	246	37	1	1.666667	...

5 rows × 29 columns

4.1.8. Find the number of trips for each zone/location ID

	LocationID	zone	borough	trip_count
0	1	Newark Airport	EWR	53
1	2	Jamaica Bay	Queens	0
2	3	Allerton/Pelham Gardens	Bronx	2
3	4	Alphabet City	Manhattan	340
4	5	Arden Heights	Staten Island	0

The next step is creating a color map (choropleth map) showing zones by the number of trips taken.

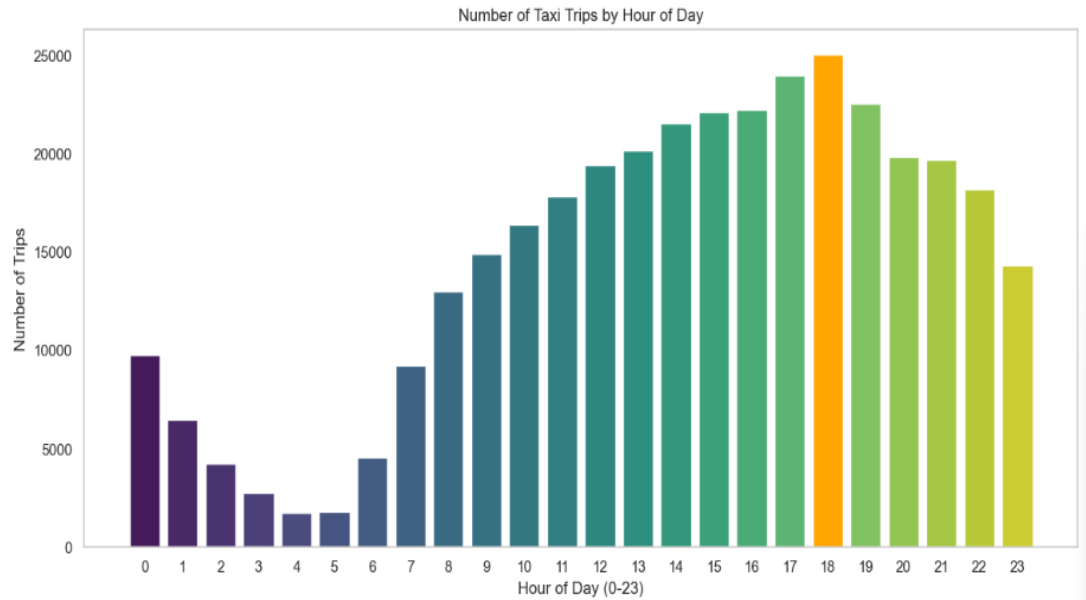
4.1.9.

```
sorted_zones = zones_trips.sort_values('trip_count', ascending=False)

# display top 20 zones by trip count
sorted_zones[['LocationID', 'zone', 'borough', 'trip_count']].head(15)
```

[54]:		LocationID	zone	borough	trip_count
131	132		JFK Airport	Queens	19417
236	237		Upper East Side South	Manhattan	16791
160	161		Midtown Center	Manhattan	16349
235	236		Upper East Side North	Manhattan	15034
137	138		LaGuardia Airport	Queens	12808
161	162		Midtown East	Manhattan	12702
185	186		Penn Station/Madison Sq West	Manhattan	12137
229	230		Times Sq/Theatre District	Manhattan	11640
141	142		Lincoln Square East	Manhattan	11515
169	170		Murray Hill	Manhattan	10370
162	163		Midtown North	Manhattan	10172
238	239		Upper West Side South	Manhattan	9742
233	234		Union Sq	Manhattan	9665
47	48		Clinton East	Manhattan	9402
67	68		East Chelsea	Manhattan	9259

4.1.10. Plot a map of the zones showing number of trips



4.1.11.

4.2. Detailed EDA: Insights and Strategies

4.2.1. Identify slow routes by comparing average speeds on different routes

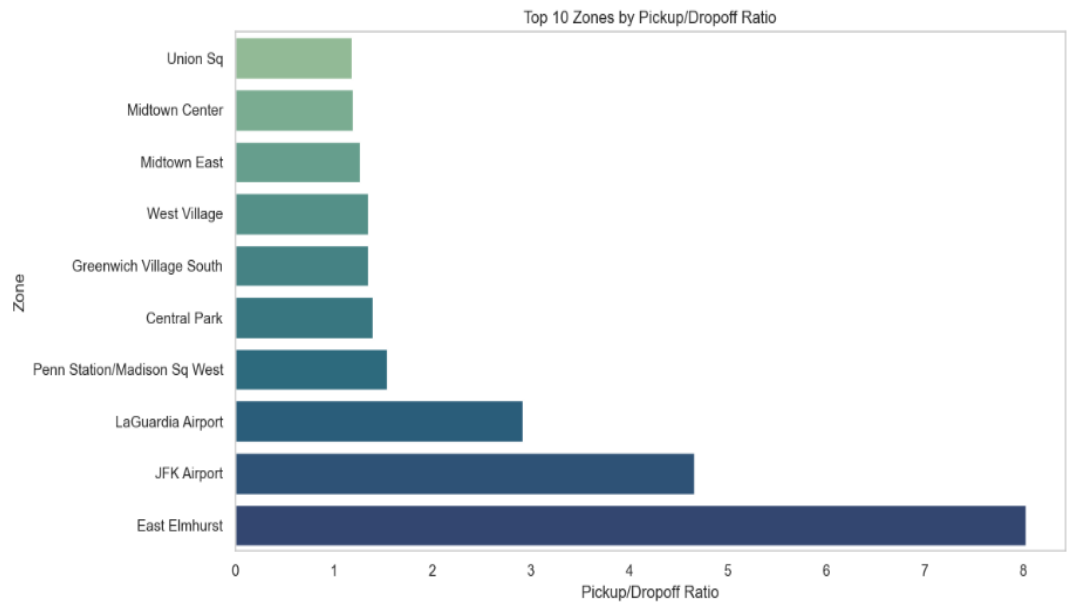
Graph Attached Here

4.2.2. Calculate the hourly number of trips and identify the busy hours

4.2.3. Scale up the number of trips from above to find the actual number of trips

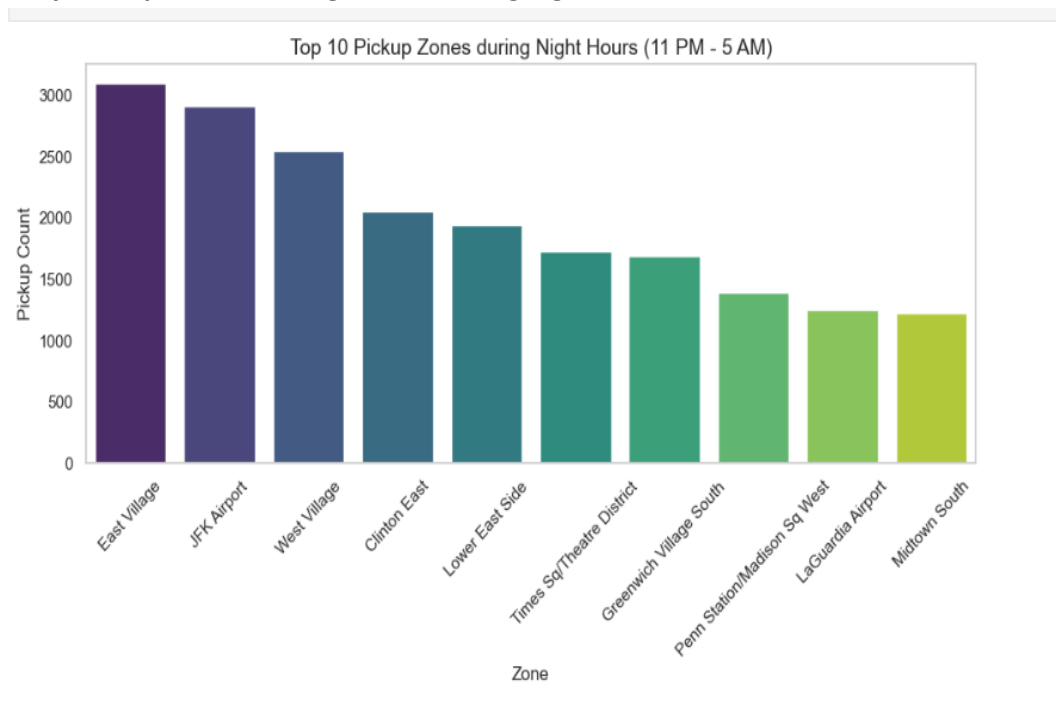
4.2.4. Compare hourly traffic on weekdays and weekends

4.2.5. Identify the top 10 zones with high hourly pickups and drops

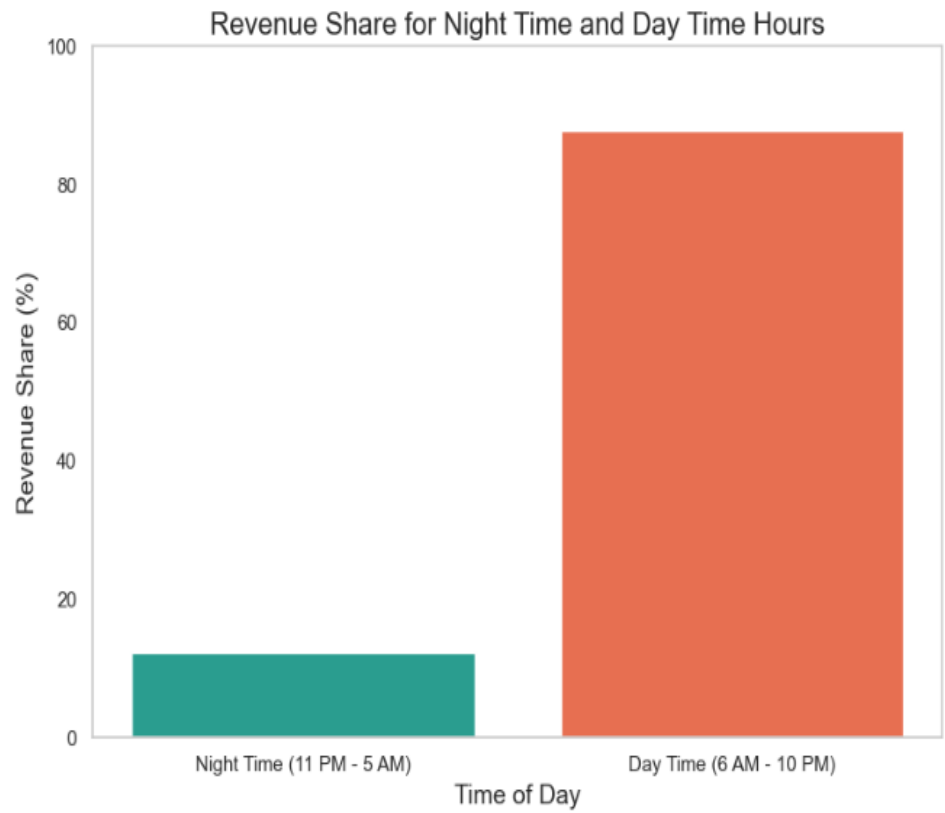


4.2.6. Find the ratio of pickups and dropoffs in each zone

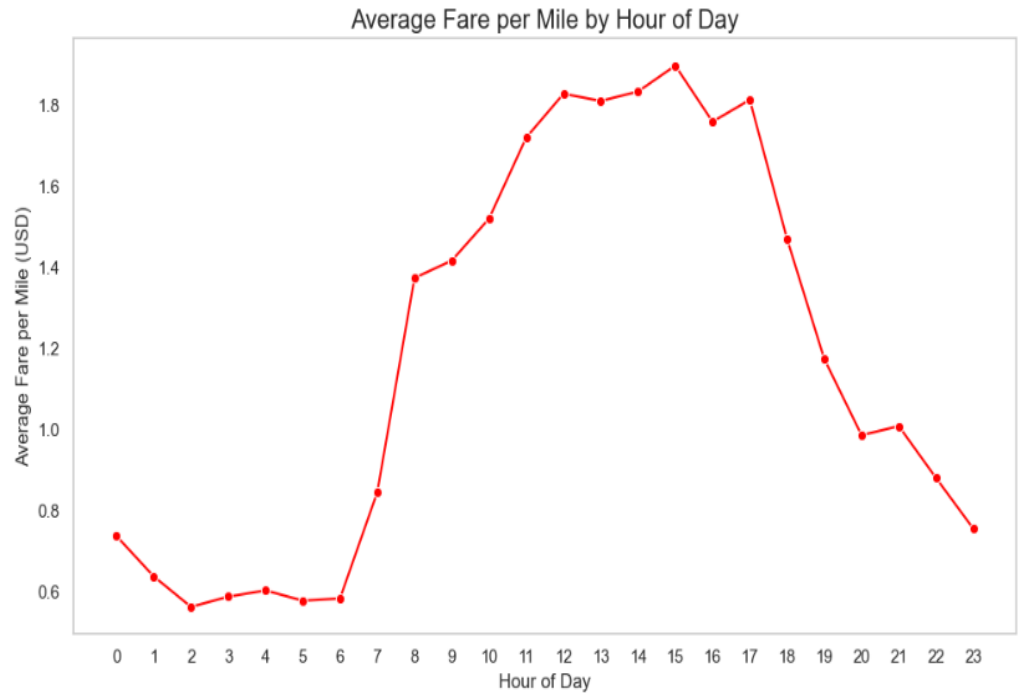
4.2.7. Identify the top zones with high traffic during night hours



4.2.8.

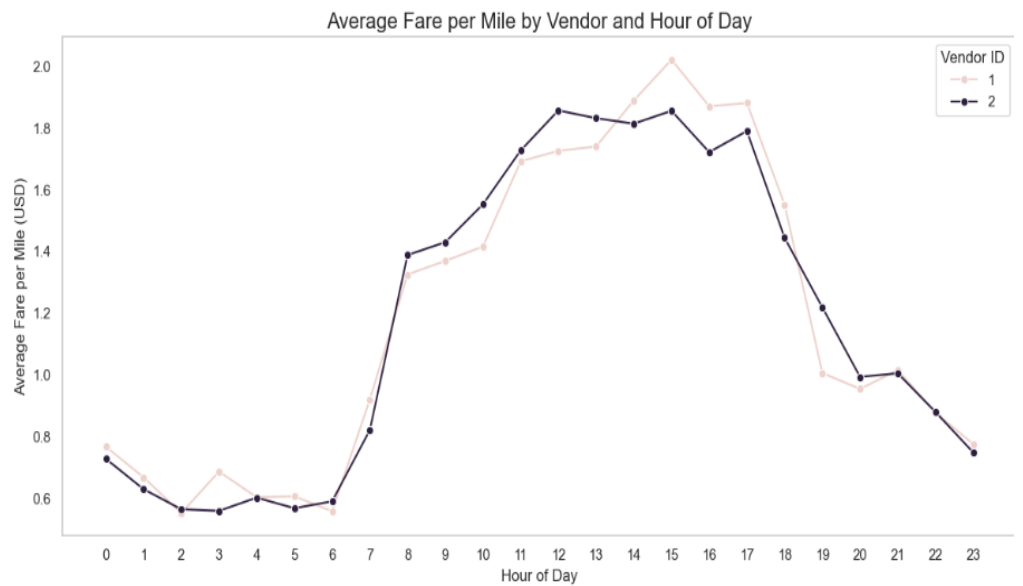


4.2.9. For the different passenger counts, find the average fare per mile per passenger



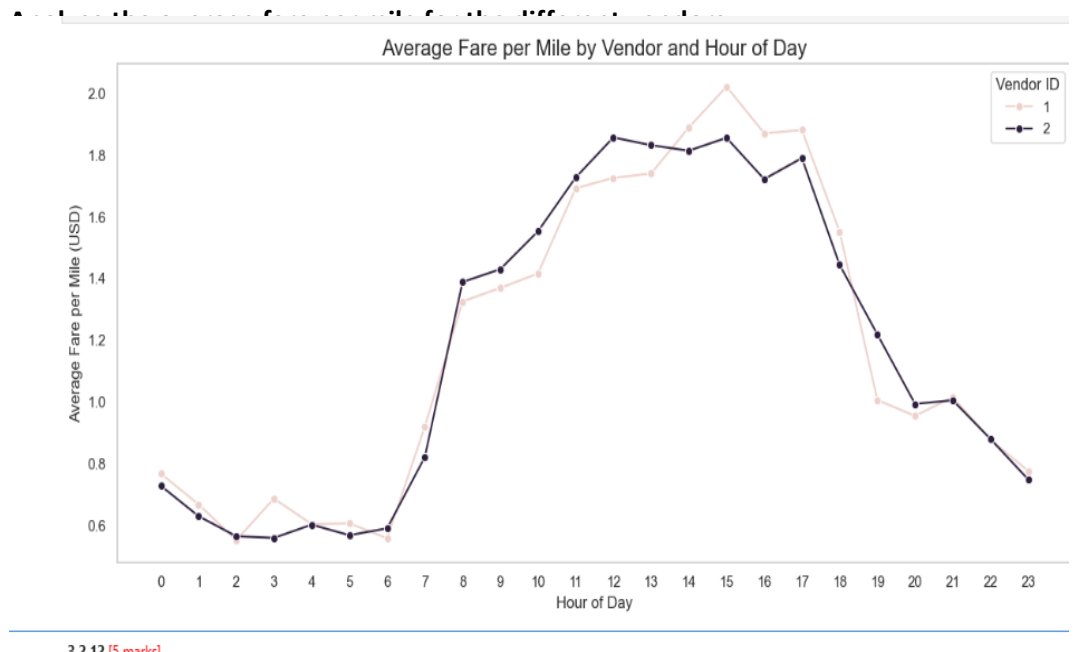
3.2.11 [3 marks]

4.2.10. Find the average fare per mile by hours of the day and by days of the week



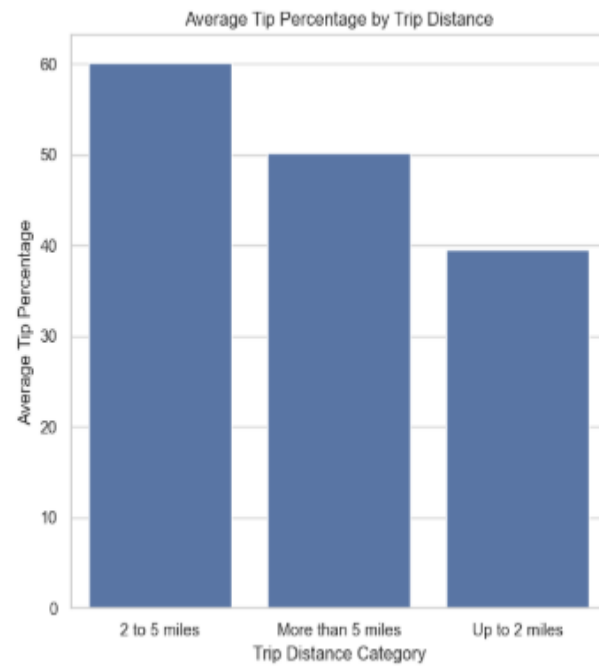
3.2.12 [5 marks]

4.2.11.

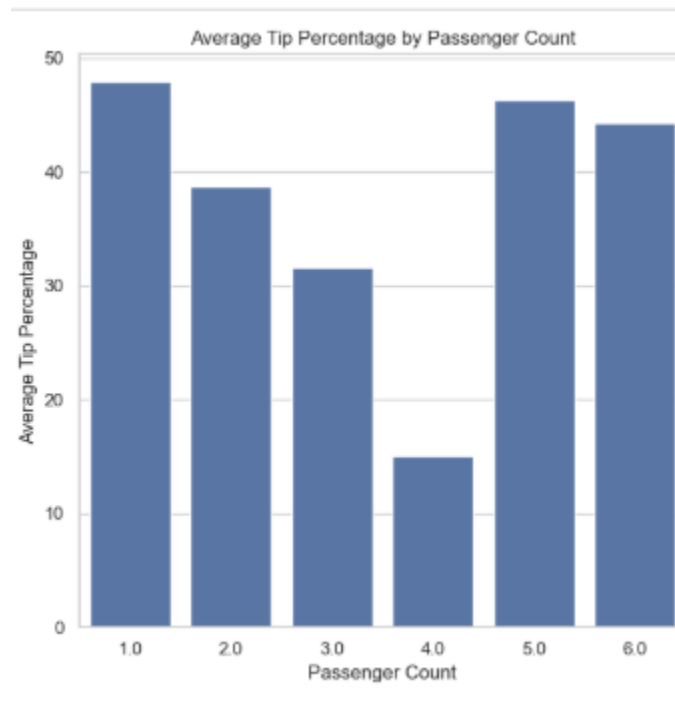


4.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

4.2.13.



4.2.14. Analyse the trends in passenger count



4.2.15. Analyse the variation of passenger counts across zones

- 4.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.**

5. Conclusions

5.1. Final Insights and Recommendations

5.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

Smart Scheduling -> Put more taxis on the road during busy times: weekday mornings (7-10AM), evenings (4-8PM), and weekend nights

Reduce taxis during slow periods to save money

Better Route Planning -> Use traffic updates to help drivers avoid busy areas like Times Square

Position drivers where rides will be needed before demand increases

Strategic Coverage -> Focus on busy pickup spots like Midtown and airports

Add more taxis in growing areas like Brooklyn during evenings

Data Cleanup -> Create better checks for unusual fares and trip distances

Look closely at areas with strange patterns to fix problems

Tailored Services -> Offer shared rides during office hours when many people travel similar routes

Provide premium options for late-night downtown customers
Reduce taxis during slow periods to save money

Better Route Planning -> Use traffic updates to help drivers avoid busy areas like Times Square

Position drivers where rides will be needed before demand increases

Strategic Coverage -> Focus on busy pickup spots like Midtown and airports

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Data Cleanup -> Create better checks for unusual fares and trip distances

Look closely at areas with strange patterns to fix problems

Tailored Services -> Offer shared rides during office hours when many people travel similar routes

Provide premium options for late-night downtown customers *****

Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

Taxi Positioning Strategy Made Simple

Best Places for Taxis

Midtown Manhattan

When: Weekday mornings (7-10AM) and evenings (4-8PM), especially Tuesday-Thursday

Why: Office workers, tourists, regular commuters

Do: Have extra taxis ready during office rush hours

Airports (JFK, LGA)

When: Every day, especially 5-11PM on weekends and holidays

Why: Travelers with luggage need taxis

Do: Match taxi availability with flight arrivals

Downtown and Financial District

When: Weekday mornings (7-9AM) and evenings (5-7PM)

Why: Business people taking short to medium trips

Do: Focus on start and end of business hours

Brooklyn and Queens

When: Evenings and weekends

Why: People going home or out for fun

Do: Move taxis from Manhattan to these areas at night and on weekends

Outer Boroughs (Bronx, Staten Island)

When: Early mornings and weekends

Why: Fewer transportation options

Do: Offer special booking options during quieter times

Time Tips

Weekdays: Focus on business areas and commuter routes

Weekends: Cover entertainment spots, airports, and neighborhoods

Seasonal Changes -> Add more taxis in tourist areas from May-September and during holiday shopping seasons

5.1.2. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

1. Match Prices to Different Customer Types

Work Travelers: These people care more about being on time than saving a few dollars. Offer reliable service during weekday commute hours.

Weekend Fun-Seekers: These customers watch their spending more closely. Give them group discount options.

Tourists: These riders want simple pricing and safe service from airports and landmarks. Offer clear flat rates.

2. Reward Regular Riders Offer monthly ride passes for frequent customers Give discounts or cashback to people who use the service often

3. Smarter Price Adjustments

Use traffic, weather, and other data to spot when rides will be difficult Add small price increases (\$1-3) only when necessary, not just because it's busy

4. Simple Flat Rates Between Popular Places Set fixed prices for common routes like airport trips Customers prefer knowing the exact fare before riding

5. Stay Competitive Without Losing Money Keep track of what Uber, Lyft and others are charging Price just below competitors, when possible, but not so low that you lose money

This approach keeps prices fair while still making profit, keeps customers coming back, and stays competitive without relying on surprise price increases.